

Comprehensive Lecture Notes

The Bargaining Problem

John F. Nash, Jr. (1950)
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Course: Game Theory & Mathematical Economics
Audience: Undergraduate / Postgraduate Students
Prepared with: Indian economic and real-world examples throughout

Learning Objectives

1. Understand the formal setup of a two-person bargaining situation.
2. Master the axioms (assumptions) Nash uses and why each is reasonable.
3. Derive and understand the **Nash Bargaining Solution** (NBS).
4. Apply the NBS to concrete barter and monetary examples.
5. Connect the theory to real Indian economic scenarios.

These notes are prepared for classroom use. All examples and interpretations are the instructor's additions to Nash's original paper.

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1 Introduction and Motivation

1.1 What is the Bargaining Problem?

Bargaining is one of the oldest and most pervasive economic activities. At its heart, it asks:

“When two parties can gain by cooperating, how should the gains be divided?”

Nash formalises this as a **two-person cooperative game** and finds a unique, axiomatically justified answer.

Sabzi Mandi (Vegetable Market) — Why Bargaining Matters

Every morning in a sabzi mandi in Delhi or Mumbai, a farmer and a shopkeeper negotiate the price of tomatoes. The farmer wants **Rs.30/kg**; the shopkeeper is willing to pay at most **Rs.40/kg** but hopes to pay **Rs.20/kg**. Any price between **Rs.30** and **Rs.40** is **mutually beneficial** — both prefer a deal to no deal.

Question: What price will they agree on? Nash’s theory gives us a principled answer.

1.2 Real-World Applications Identified by Nash

Nash lists these economic structures as bargaining problems:

Economic Situation	Description
Monopoly vs. Monopsony	Single seller negotiating with a single buyer
State Trading	Two governments negotiating trade terms
Labour–Management	Union negotiating wages with an employer
Bilateral Monopoly	Cournot, Bowley, Tintner, Fellner’s classical problem

Indian Applications

- **OPEC–India oil deals:** India (IOC) negotiates bulk crude oil prices — classic bilateral negotiation.
- **MSME supplier–OEM:** A small auto-parts manufacturer negotiating price with Maruti Suzuki.
- **Agricultural Produce Market Committees (APMCs):** Farmer unions bargaining with licensed traders.
- **FCI procurement:** Food Corporation of India setting minimum support prices — a form of mandated bargaining solution.

1.3 Connection to Game Theory

Nash notes that the bargaining problem can be viewed as a **nonzero-sum two-person game**. This is important:

Key Insight / Watch Out

In a **zero-sum game**, one player's gain equals the other's loss (e.g., cricket — every run India scores, the other team loses). In a **nonzero-sum game**, *both* players can gain simultaneously. Bargaining is fundamentally nonzero-sum.

The reference work is von Neumann & Morgenstern's *Theory of Games and Economic Behavior* (1944), but that book did not find values for n -person games. Nash fills this gap.

2 Utility Theory of the Individual

Before building the bargaining model, Nash reviews **von Neumann–Morgenstern expected utility theory**. This is the mathematical language for expressing preferences.

2.1 The Concept of an Anticipation

Anticipation

An **anticipation** is a state of expectation — a probability distribution over future outcomes. If an individual has probability p of outcome A and probability $(1 - p)$ of outcome B , we write this anticipation as:

$$pA + (1 - p)B, \quad 0 \leq p \leq 1.$$

IPL Ticket Lottery

Suppose you enter a lucky draw:

- With probability $\frac{1}{2}$, you win a front-row IPL final ticket (outcome A).
- With probability $\frac{1}{2}$, you win nothing (outcome B).

Your anticipation is $\frac{1}{2}A + \frac{1}{2}B$.

Nash's theory says we can assign a *single number* (utility) to this anticipation that captures how much you value it.

2.2 The Five Axioms for Individual Utility

Nash assumes five properties about individual preferences:

Axiom 1 — Completeness (Comparability)

An individual offered two possible anticipations can decide which is preferable, or that they are equally desirable.

In plain English: You can always compare any two lotteries. No “I can’t decide at all”.

Axiom 2 — Transitivity

The ordering is transitive: if $A \succ B$ and $B \succ C$, then $A \succ C$.

In plain English: Preferences are consistent. If you prefer Swiggy over Zomato and Zomato over Dunzo, you must prefer Swiggy over Dunzo.

Axiom 3 — Equal Desirability of Probability Mixtures

Any probability combination of equally desirable states is just as desirable as either.

If $A \sim B$, then for any $0 \leq p \leq 1$: $pA + (1 - p)B \sim A \sim B$.

Axiom 4 — Continuity

If $A \succ B \succ C$, then there exists a $p \in [0, 1]$ such that:

$$B \sim pA + (1 - p)C.$$

In plain English: No outcome is so catastrophic (or so magnificent) that no probability mixture can match any intermediate outcome. This is an assumption of “no lexicographic preferences.”

Axiom 5 — Independence / Substitutability

If $A \sim B$, then for any C and $0 \leq p \leq 1$:

$$pA + (1 - p)C \sim pB + (1 - p)C.$$

Moreover, A may be substituted for B in any preference ordering.

In plain English: If you’re indifferent between chai and coffee, substituting one for the other in any mixture doesn’t change your overall preference.

Key Insight / Watch Out

Axiom 5 (Independence) is the most controversial axiom in utility theory. It is violated in the famous **Allais Paradox** (1953). For this course, we accept it as Nash did.

2.3 The Utility Function

Existence of Utility Function

These five axioms guarantee the existence of a function $u : \mathcal{A} \rightarrow \mathbb{R}$ (where $\mathcal{A}$ is the set of all anticipations) satisfying:

$$(a) \quad u(A) > u(B) \iff A \succ B \text{ (and } u(A) = u(B) \iff A \sim B).$$

(b) **Linearity (expected utility property):**

$$u[pA + (1-p)B] = pu(A) + (1-p)u(B), \quad \forall 0 \leq p \leq 1.$$

Uniqueness Up to Positive Affine Transformation

The utility function is **not unique**. If u is a valid utility function, so is:

$$\tilde{u} = au + b, \quad a > 0, b \in \mathbb{R}.$$

This is called a **positive affine transformation**. It changes the scale and origin but preserves preference orderings and the linearity property.

Temperature Analogy for Affine Invariance

Just as temperature in Celsius and Fahrenheit measure the same thing ($F = 1.8C + 32$), utility functions u and $\tilde{u} = au + b$ represent the same preferences. No specific scale is privileged — only *comparisons* matter.

3 The Two-Person Bargaining Model

3.1 Setup: Two-Person Anticipations

Now Nash extends utility theory to two players.

Two-Person Anticipation

A **two-person anticipation** $[A, B]$ pairs one-person anticipations: A for individual 1, B for individual 2. A probability mixture of two such pairs is:

$$p[A, B] + (1-p)[C, D] = [pA + (1-p)C, pB + (1-p)D].$$

3.2 The Status Quo (Disagreement Point)

Disagreement Point / Status Quo

In any bargaining situation, there is a distinguished anticipation: **no cooperation**. This is what happens if negotiations break down. Nash assigns utility zero to this point for both players:

$$u_1(\text{no deal}) = 0, \quad u_2(\text{no deal}) = 0.$$

This normalises the utility functions without loss of generality (we can always re-label).

NREGA Wage Negotiation

Consider a daily wage labourer in rural Rajasthan negotiating with a landlord.

- **Disagreement point for labourer:** Falls back to NREGA wages (Rs.220/day guaranteed).
- **Disagreement point for landlord:** Must hire at market rate or leave crop unharvested.

Nash sets utility at the disagreement point equal to zero, so all utilities represent *gains above the fallback*.

3.3 The Feasible Set S

Feasible Set

Plot the utility pairs (u_1, u_2) for *all* available anticipations. The resulting set $S \subset \mathbb{R}^2$ is the **feasible set** (or set of alternatives).

Nash imposes two geometric assumptions on S :

Compactness of S

S is **compact** (closed and bounded). This ensures a maximum exists and that the feasible utility pairs do not go to infinity.

Convexity of S

S is **convex**: for any two points $(u_1, u_2), (v_1, v_2) \in S$ and $0 \leq p \leq 1$:

$$(p u_1 + (1 - p) v_1, p u_2 + (1 - p) v_2) \in S.$$

Reason: If anticipation X yields (u_1, u_2) and anticipation Y yields (v_1, v_2) , then the *lottery* “ X with probability p , Y with probability $1 - p$ ” yields the mixed point. Lotteries are always available, so S must be convex.

Geometric Intuition Using a Land Deal

Imagine a farmer and a builder negotiating over a plot. The farmer values the land as farmland (utility pair A), the builder values it as housing (utility pair B). Any *partial deal* — say, 60% farmland, 40% housing — is also feasible. Convexity captures this.

3.4 The Solution Function

Nash seeks a **solution function** $c(S)$ that maps each feasible set S (containing the origin) to a unique point $(u_1^*, u_2^*) \in S$ — the *recommended bargaining outcome*.

We also restrict to cases where there is some $(u_1, u_2) \in S$ with $u_1 > 0$ and $u_2 > 0$ (both can benefit — otherwise there's no real bargaining problem).

4 Nash's Axioms for the Bargaining Solution

Nash imposes three axioms on the solution function $c(S)$.

Axiom 6 — Pareto Optimality (Efficiency)

If $\alpha \in S$ and there exists $\beta \in S$ such that $u_1(\beta) > u_1(\alpha)$ and $u_2(\beta) > u_2(\alpha)$, then $\alpha \neq c(S)$.

In plain English: The solution must be **Pareto optimal** — there should be no other outcome that makes *both* parties better off. Rational bargainers will never leave “money on the table.”

Trade Union–Management Negotiation

Suppose the current proposal gives workers **Rs.15,000/month** and the company profits **Rs.50 lakh**. If another deal exists giving workers **Rs.16,000** AND company profits **Rs.52 lakh** — the first deal is *not* Pareto optimal. Axiom 6 says the solution cannot be at the first deal.

Axiom 7 — Independence of Irrelevant Alternatives (IIA)

If $S \subseteq T$ and $c(T) \in S$, then $c(T) = c(S)$.

In plain English: If the solution chosen from a *larger* feasible set T happens to lie inside a *smaller* set S , then restricting the alternatives to S should not change the solution.

Nash's Intuition for IIA

Suppose two rational parties agree that outcome $c(T)$ is fair when all of T is available. They then agree to restrict attention to outcomes in $S \subset T$ (perhaps by pre-committing not to demand points outside S). Since $c(T) \in S$, they would still agree on $c(T)$. So $c(S) = c(T)$.

Axiom 8 — Symmetry

If S is **symmetric** — meaning whenever $(a, b) \in S$ we have $(b, a) \in S$ — then the solution satisfies $c(S) = (a^*, a^*)$ for some a^* , i.e., both players receive equal utility.

In plain English: If the situation is perfectly symmetric (both parties have the same bargaining power, same preferences, same alternatives), the solution should treat them equally.

Key Insight / Watch Out

Nash explicitly assumes the two bargainers are **equal in bargaining skill**. This is why Axiom 8 (symmetry) is appropriate here. If skills were unequal, we would use the **Asymmetric Nash Bargaining Solution** (not covered in the original paper).

5 The Main Result: Nash Bargaining Solution

5.1 Statement of the Theorem

Nash Bargaining Theorem

There is a **unique** solution function $c : \mathcal{S} \rightarrow \mathbb{R}^2$ satisfying Axioms 6, 7, and 8. It is:

$$c(S) = \arg \max_{(u_1, u_2) \in S, u_1 \geq 0, u_2 \geq 0} u_1 \cdot u_2$$

The solution is the point in the **first quadrant** of S that *maximises the product of utilities*.

This is the famous **Nash Bargaining Solution (NBS)**.

5.2 Proof Sketch (Following Nash)

Step 1. Existence and Uniqueness of the Maximiser. Since S is compact, the continuous function $f(u_1, u_2) = u_1 u_2$ attains a maximum on $S \cap \{u_1 \geq 0, u_2 \geq 0\}$. Since S is convex, this maximum is unique (the product function $u_1 u_2$ is strictly quasi-concave in the first quadrant).

Step 2. Normalisation. Re-scale the utility functions (positive affine transformations are allowed) so that the maximiser moves to the point $(1, 1)$. Call this transformed set $\tilde{S}$.

Step 3. Claim: No point in $\tilde{S}$ has $u_1 + u_2 > 2$. If a point $P = (u_1^P, u_2^P)$ satisfied $u_1^P + u_2^P > 2$, then somewhere on the segment from $(1, 1)$ to P there would be a point with $u_1 u_2 > 1$, contradicting that $(1, 1)$ is the maximiser. (The AM–GM inequality gives $u_1 u_2 \leq \left(\frac{u_1 + u_2}{2}\right)^2$, with equality iff $u_1 = u_2$. If $u_1 + u_2 > 2$ and $u_1 \neq u_2$, a better product exists.)

Step 4. Construct a symmetric enclosing square. Build a square T in the region $u_1 + u_2 \leq 2$, symmetric about the line $u_1 = u_2$, with one side on $u_1 + u_2 = 2$, and enclosing $\tilde{S}$.

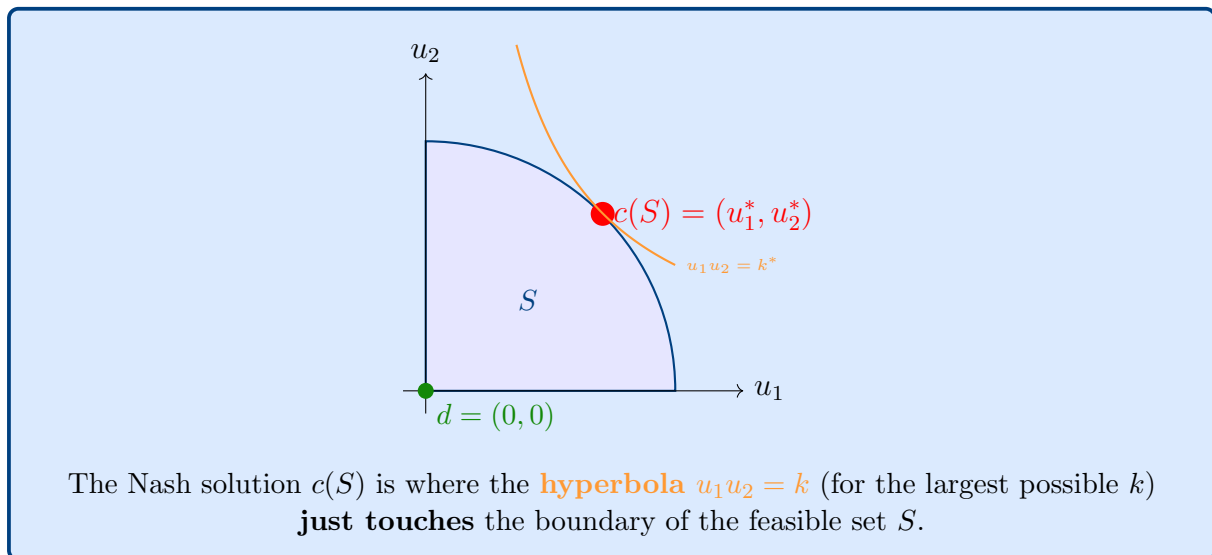
Step 5. Apply Symmetry (Axiom 8) to the square. For the square T , by symmetry and Pareto optimality, the only acceptable solution on the boundary $u_1 + u_2 = 2$ that is symmetric is $(1, 1)$. Hence $c(T) = (1, 1)$.

Step 6. Apply IIA (Axiom 7). Since $\tilde{S} \subseteq T$ and $c(T) = (1, 1) \in \tilde{S}$, by Axiom 7:

$$c(\tilde{S}) = (1, 1).$$

Step 7. Conclusion. The unique solution is the point maximising $u_1 u_2$ in S . □

5.3 Geometric Interpretation



6 Examples from the Paper

6.1 Example 1: Barter Without Money (Bill and Jack)

Nash gives a concrete barter example. Two people, Bill and Jack, trade goods. Here is the utility table:

Good (Owner)	Utility to Bill	Utility to Jack
<i>Bill's goods:</i>		
Book	2	4
Whip	2	2
Ball	2	1
Bat	2	2
Box	4	1
<i>Jack's goods:</i>		
Pen	10	1
Toy	4	1
Knife	6	2
Hat	2	2

6.1.1 How to Compute the Nash Bargaining Solution Here

- Initial endowments:** Bill currently holds {book, whip, ball, bat, box} with total utility to Bill = $2 + 2 + 2 + 2 + 4 = 12$ and to Jack = $4 + 2 + 1 + 2 + 1 = 10$. Jack holds {pen, toy, knife, hat} with utility to Bill = $10 + 4 + 6 + 2 = 22$ and to Jack = $1 + 1 + 2 + 2 = 6$.
- Find the trade that maximises** $u_1 \cdot u_2$ (where utilities are measured as gains above no-trade).
- Nash's answer:**
 - Bill gives Jack: {book, whip, ball, bat} (Jack gains $4 + 2 + 1 + 2 = 9$ utility, Bill loses $2 + 2 + 2 + 2 = 8$)
 - Jack gives Bill: {pen, toy, knife} (Bill gains $10 + 4 + 6 = 20$ utility, Jack loses $1 + 1 + 2 = 4$)
- The resulting utility gains: $u_1^* =$ (Bill's net gain), $u_2^* =$ (Jack's net gain), and their product is maximised.

Key Insight / Watch Out

The solution lies at a **vertex** of the convex polygon (feasible set) because utilities are integers and the feasible set is a polygon. At the solution, there is exactly one anticipation (one specific trade) realising it.

6.2 Example 2: Bargaining With Money

When a **common medium of exchange** (money) is available:

- The **money equivalent** of a good = amount of money equally desirable to the individual.
- When utility is approximately **linear in money** (valid for small amounts), money utility functions are usable directly.

- The feasible set in the first quadrant becomes an **isosceles right triangle**.

Nash Solution With Money

When both parties have linear utility in money, the Nash bargaining solution gives each party the **same money profit**.

$$\Delta m_1^* = \Delta m_2^*$$

Splitting a Joint Venture Profit

Tata and a foreign partner form a joint venture that generates **Rs.100** crore in profit above both parties' outside options. The Nash solution (with symmetric, linear money utility) splits this **50–50**: each party gets **Rs.50** crore.

Important: This is a split of *surplus above the disagreement point*, not the total revenue. If Tata's outside option was worth **Rs.30** crore and the partner's was **Rs.20** crore, Nash says they split the **Rs.50** crore surplus equally: Tata gets **Rs.30** + **Rs.25** = **Rs.55** crore; partner gets **Rs.20** + **Rs.25** = **Rs.45** crore.

7 Deep Conceptual Analysis

7.1 Why Maximise the Product? Intuition

The product $u_1 \cdot u_2$ arises naturally from the axioms. Here is economic intuition:

1. **Efficiency (Axiom 6):** The solution must be on the Pareto frontier — the “northeast boundary” of S .
2. **Symmetry (Axiom 8):** On the Pareto frontier, the solution is the point equidistant (in utility terms) from both axes — which the product captures.
3. **IIA (Axiom 7):** The product is the unique function consistent with not changing the solution when irrelevant alternatives are removed.

Together, these three properties single out the product uniquely.

7.2 The Role of the Disagreement Point

Disagreement Point Determines Bargaining Power

The disagreement point $d = (d_1, d_2)$ (Nash normalises it to $(0, 0)$) is crucial. In general:

$$c(S, d) = \arg \max_{(u_1, u_2) \in S, u_i \geq d_i} (u_1 - d_1)(u_2 - d_2).$$

A party with a **better outside option** (higher d_i) gets a **better Nash outcome**. This is why BATNA (Best Alternative to a Negotiated Agreement) is so important in real negotiations.

BATNA in Indian Job Negotiations

A software engineer with a competing offer from TCS for **Rs.12 LPA** is negotiating with Infosys. Her BATNA (disagreement point) is **Rs.12 LPA**. Nash theory says: the higher your BATNA, the more you can extract from the negotiation. This is precisely why candidates with competing offers get better packages!

7.3 Assumptions and Limitations

Critical Assumptions and Their Limitations

1. **Perfect rationality:** Both parties are fully rational, can compute utilities exactly. Real bargainers have cognitive limitations (bounded rationality — Herbert Simon).
2. **Equal bargaining skill:** Not realistic. Unequal power (e.g., a monopsonist employer vs. a single worker) requires the **Asymmetric Nash Solution:** $\max(u_1 - d_1)^\alpha (u_2 - d_2)^{1-\alpha}$ where α captures relative power.
3. **Full information:** Each party knows the other's utility function. In practice, information is private and asymmetric (this is handled by Harsanyi's model).
4. **Static / single-shot:** The model doesn't address how bargaining *unfolds over time*. Rubinstein (1982) extended Nash's axiomatic solution to a dynamic alternating-offers model.
5. **Transferable utility assumed in money case:** The split-the-surplus result requires linear utility in money, which breaks for risk-averse parties.

8 Extensions and Related Concepts

8.1 Asymmetric Nash Bargaining Solution

When bargaining powers differ (weights α and $1 - \alpha$):

$$c_\alpha(S, d) = \arg \max_{u \in S, u \geq d} (u_1 - d_1)^\alpha (u_2 - d_2)^{1-\alpha}$$

Unequal Joint Ventures

In many India–foreign joint ventures under old FEMA rules, the Indian partner was limited to a minority stake. This is economically equivalent to $\alpha < \frac{1}{2}$ for the foreign firm in the Nash bargaining framework.

8.2 Kalai–Smorodinsky Solution

An alternative to Nash that replaces IIA with a monotonicity axiom:

The solution is on the segment from d to the “ideal point” $(u_1^{\max}, u_2^{\max})$.

8.3 Rubinstein's Alternating Offers Model (1982)

Rubinstein showed that if two parties alternate making offers (with discounting), the *unique subgame-perfect equilibrium* converges to the Nash bargaining solution as discounting vanishes. This provides a **non-cooperative foundation** for Nash's cooperative axioms.

9 Summary and Formula Sheet

Complete Summary

Problem: Two rational, equally skilled bargainers with utility functions u_1, u_2 face feasible set S (compact, convex) with disagreement point $d = (0, 0)$.

Three axioms on solution $c(S)$:

1. **Pareto Optimality:** No feasible alternative is strictly better for both.
2. **IIA:** Removing unchosen alternatives doesn't change the solution.
3. **Symmetry:** Symmetric problems have equal-utility solutions.

Unique solution satisfying all three:

$$c(S) = \arg \max_{(u_1, u_2) \in S, u_1 \geq 0, u_2 \geq 0} u_1 \cdot u_2$$

With non-zero disagreement point $d = (d_1, d_2)$:

$$c(S, d) = \arg \max_{u \in S, u \geq d} (u_1 - d_1)(u_2 - d_2)$$

Special case — linear money utility:

$$\Delta m_1^* = \Delta m_2^* \quad (\text{equal split of the surplus})$$

9.1 Common Exam Questions

1. Define a bargaining problem in Nash's framework. What conditions must the feasible set S satisfy?
2. State and explain Nash's three axioms for the bargaining solution. Give an economic justification for each.
3. Prove that the Nash Bargaining Solution maximises the product $u_1 \cdot u_2$.
4. Two firms can jointly earn **Rs.1000** crore. Firm A's outside option yields **Rs.200** crore; Firm B's yields **Rs.300** crore. Find the Nash Bargaining Solution.

$$\text{Surplus} = 1000 - 200 - 300 = 500 \text{ crore}$$

$$c = \left(200 + \frac{500}{2}, 300 + \frac{500}{2} \right) = (450, 550) \text{ crore}$$

5. What is the role of the disagreement point in determining bargaining power?
6. Explain why the feasible set must be convex in Nash's model.
7. What is the Independence of Irrelevant Alternatives axiom? Give an example where it might be violated.

10 Glossary of Key Terms

Term	Definition
Anticipation	A probability distribution over future outcomes (a lottery).
Utility Function	A real-valued function representing preferences, linear in probabilities.
Feasible Set S	The set of all utility pairs achievable through bargaining.
Disagreement Point d	Utilities if no agreement is reached; the “fallback” position.
Pareto Optimal	No other outcome makes all parties weakly better and at least one strictly better.
Nash Product	$u_1 \cdot u_2$ (or $(u_1 - d_1)(u_2 - d_2)$) — the function maximised by NBS.
IIA	Independence of Irrelevant Alternatives: removing unchosen options doesn't change the solution.
Symmetry Axiom	Equal bargaining skill $\Rightarrow$ equal utility in symmetric problems.
BATNA	Best Alternative to Negotiated Agreement — the disagreement point in practice.
Affine Transformation	$\tilde{u} = au + b, a > 0$: preserves preference ordering and utility linearity.

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